CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Desiree Kamerer Fuentes 11/07/2026
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	[R03, R14]
Q02	Work order is unique	2	10%	[R05, R06]
Q03	Status uses the agreed list	2	10%	[R08, R16]
Q04	Duration is in range	3	15%	[R10, R18, R20]
Q05	READY identifies an inspector	3	15%	[R03,R07,R12]
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

While reviewing the Raw and Audit sheets, I found that Row R03 was flagged for Q01 and Q05. The Machine_ID is missing, and the status is READY even though there is no Inspector_ID assigned. This matters because the record is marked READY, but important information is missing. The spreadsheet cannot confirm which machine was inspected or who completed the inspection.

Next action: what to verify, who owns it, and how to retest

I would have the Maintenance Records Owner check the original inspection record to verify the missing Machine_ID for R03. The Shift Supervisor should also verify the missing Inspector_ID because the status is READY. Once the correct information is confirmed and updated, I would refresh Power Query and check Q01 and Q05 again to make sure the record is no longer flagged.

Limit: why you cannot guess a correction or prove accuracy

I cannot assume what the missing information should be because the spreadsheet only shows what was recorded. I have not inspected the actual machines or reviewed the original inspection documents. Even if the data passes all five rules, that does not prove the machine is safe or that the information is accurate.

Professional connection: one framework idea and how you used it

I used the data quality idea of completeness to check whether important information was missing. For example, Q01 checks for missing Machine_ID values, while Q05 checks for missing Inspector_ID values when a record says READY. These checks helped me identify records that need to be reviewed instead of assuming the information is correct.

Refresh test: before, changed value, after, and restored result

I tested the refresh by changing R01's Inspection_Minutes from 20 to 0 in the Raw sheet. Before the change, Q04 showed OK. After clicking Refresh All, Q04 changed to FLAG because 0 is outside the allowed range of 1–60 minutes. I then changed the value back to 20 and refreshed again. Q04 returned to OK. This showed that the audit updates when the source data changes, but it cannot prove that the original inspection information is accurate.

Export only Report: two landscape pages, page break before row 26. Review both pages.